

STUDENT NUMBER: _____

STUDENT NAME: _____



THE HILLS GRAMMAR SCHOOL

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

2008

MATHEMATICS

2 UNIT ADVANCED

Teacher Responsible:

Mr K McClure

General Instructions:

- Reading time - 5 minutes
- Working time - 3 hours
- Write using blue or black pen.
- Marks may be deducted for careless, untidy or badly arranged work.
- Board approved calculators may be used.
- A table of standard integrals is supplied at the back of this paper.
- ALL necessary working should be shown in every question.
- **Start each question in a new booklet.**

Total marks - 120

- Attempt Questions 1-10
- All questions are of equal value.

Students are advised that this is a Trial Examination only and cannot in any way guarantee the content or the format of the Higher School Certificate Examination.

Question 1 (12 Marks)**Marks**

- a) Solve the equation

$$4(x - 2) - 3(x - 5) = 17$$

2

- b) Calculate the sum to infinity of the series

$$2 + 1 + \frac{1}{2} + \frac{1}{4} +$$

2

- c) Solve the equation $2x^2 + 7x - 15 = 0$

2

- d) Find the perpendicular distance of the point $(10, -2)$ from the line $3x + 4y - 7 = 0$

2

- e) Rationalise the denominator of $\frac{1}{2\sqrt{2} - 1}$

2

- f) The sector of a circle of radius 4 cm subtends an angle of $\frac{5\pi}{18}$ radians at the centre. Find the arc-length subtended by this angle.
Give your answer to 1 decimal place.

2

Question 2 (12 Marks)**Marks**

a) Differentiate with respect to x and fully simplify your answer.

i) $y = x^2 e^x$ 2

ii) $y = \frac{\sin x}{\cos x}$ 2

b)

i) Find $\int \frac{x^2 + 1}{x^3 + 3x} dx$ 2

ii) Find the area under the curve $f(x) = \frac{2}{1-3x}$ from $x = 2$ to $x = 1$
Leave your answer in exact form. 3

c) The equation of the curve C is given by $y = \frac{4}{x}$. Find the equation
of the tangent to the curve C at the point $P(1, 4)$ 3

Question 3 (12 Marks)**Marks**

- a) i) Plot the points $A(3, 3)$, $B(8, 0)$, $C(-1, 1)$ and $D(-6, 4)$ **1**
- ii) Find the co-ordinates of the point of intersection of the lines AC and BD . **4**
- b) The first four terms of an arithmetic progression are 5, 11, 17 and 23.
Find i) the 30th term **1**
- ii) the sum of the first 30 terms. **2**
- c) The sum to infinity of a geometric progression is 7 and the sum of the first two terms is $\frac{48}{7}$.
i) Show that the common ratio, r , satisfies the equation $1 - 49r^2 = 0$ **2**
- ii) Find the first term of the geometric progression with a positive common ratio. **2**

Question 4 (12 Marks)**Marks**

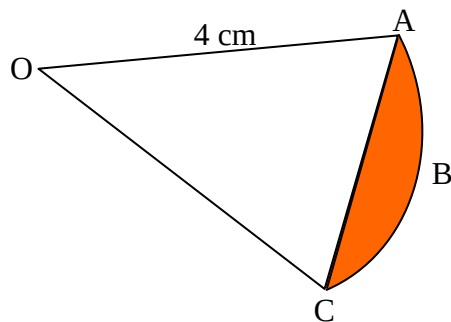
a) Solve $\cos^2 \theta = \frac{1}{2}$ for θ where $0^\circ \leq \theta \leq 360^\circ$

3

b) Solve the equation $x^4 + 5x^2 - 14 = 0$
(hint, let $m = x^2$)

3

c)



The diagram shows part of a circle, centre O and radius 4 cm. Given that the arc length ABC is 5 cm, calculate

i) The size of angle AOC in radians

2

ii) The area of the shaded region

2

d) Find the equation of the tangent to the curve $f(x) = \frac{1}{x^2}$
at the point $P(-1, 1)$

2

Question 5 (12 Marks)

Marks

- a) Find the values of the constant k given that the equation $(5k + 1)x^2 - 8kx + 3k = 0$ has equal roots. **3**
- b) Find the maximum area of a rectangle which has a perimeter of 28 units by first constructing a formula for area (A) in terms of the length of one side (x). **3**
- c) Determine the intervals for which the function $f(x) = x^4 - 8x^2 - 3$ is increasing and decreasing. **3**
- d) Find the area enclosed between the curve $y = x^2 - 2x - 3$ and the line $y = x + 1$ **3**

Question 6 (12 Marks)**Marks**

- a) Solve the equation $3^x = 10$ for x (to 1 decimal place). **2**
- b) Consider the curve given by $f(x) = x^4 - 4x^3$.
- i) find the stationary points **2**
- ii) determine the nature of the stationary points **2**
- iii) find any points of inflexion **1**
- iv) sketch the curve clearly showing all relevant points. **2**
- c) Find the centre and the radius of the circle $x^2 + y^2 + 2x - 4y - 4 = 0$ **3**

Question 7 (12 Marks)**Marks**

- | | | | |
|----|---|--|----------|
| a) | i) | Express $y = x^2 + 6x + 6$ in the form $(x-h)^2 = 4a(y-k)$ | 2 |
| | ii) | state the vertex | 1 |
| | iii) | state the focus | 1 |
| | iv) | find the equation of the directrix | 1 |
| b) | The area enclosed between the curve $y = 4 - x^2$ and the line $y = 4 - 2x$ is rotated through 2π radians about the x-axis. Find the volume of the solid generated, leave your answer in terms of π . | | 4 |
| c) | Consider the equation $y = 3\cos 2\theta$ for θ where $0^\circ < \theta < 360^\circ$ | | |
| | i) | state the amplitude of the curve | 1 |
| | ii) | find the period of the curve | 1 |
| | iii) | Make a neat sketch of the curve | 1 |

Question 8 (12 Marks)**Marks**

- a) Find the equation of the normal to the curve $y = \ln\left(\frac{x-1}{x+1}\right)$ at the point P where $x = 3$ **3**
- b) Using Simpson's rule with 5 ordinates, find an approximation for the area under the curve $f(x) = e^{-2x}$ between $x=1$ and $x = 3$ **3**
- c) Solve the equation $4\cos^2 \theta + 3\sin \theta = 4$ where $0^\circ \leq \theta \leq 360^\circ$ (Hint: let $\cos^2 \theta = 1 - \sin^2 \theta$) **3**
- d) Find $\frac{dy}{dx}$ given that $y = \sqrt{6+x}\sqrt{3-2x}$ **3**

Question 9 (12 Marks)**Marks**

- a) A particle moves in a straight line and at any time t seconds ($t \geq 0$), its displacement from a fixed point 0 on the line is given by $x(t) = 4 + 2 \cos \frac{\pi}{5}t$, $0 \leq t \leq 20$

i) Sketch the graph of the displacement against time.

3

ii) Find when the velocity of the particle is zero.

3

- b) The population $P(t)$ of persons in a new suburb increases at a rate given by the equation $\frac{dP}{dt} = kP$, where k is a constant and t is the time in years. The population of the suburb is expected to double every fifteen years.

It may be assumed that $P = P_0 e^{kt}$ where P_0 is the initial population.

i) Find the value of k . Leave your answer in exact form.

2

ii) In which year will the suburb attain a population four times that which it had at the beginning of the year 2000?

2

iii) It is known the population of the suburb at the beginning of the year 2000 was 30,000. What is the population expected to be in the year 2035?

2

Question 10 (12 Marks)**Marks**

- a) A piece of wire of length 1 m is cut into two parts and each part is bent to form a square. If the total area of the two squares formed is 325 cm^2 , find the perimeter of each square.

6

- b) A closed, right circular cylinder of base radius r cm and height h cm has a volume of $54\pi \text{ cm}^3$.

- i) Show that S , the total surface area of the cylinder is given by

$$S = \frac{108\pi}{r} + 2\pi r^2$$

3

- ii) Hence find the radius and height which make the surface area minimum.

3**END OF EXAMINATION**

Q1a) $4x - 8 - 3x + 15 = 17$

$x + 7 = 17$

$x = 10$

b) $a = 2, r = \frac{1}{2} \quad S_{\infty} = \frac{a}{1-r} = \frac{2}{\frac{1}{2}} = \underline{\underline{4}}$

c) $(2x-3)(x+5) = 0 \quad x = 3/2, -5$

d) $d = \frac{|30-8-7|}{\sqrt{3^2+4^2}} = \frac{15}{5} = 3$

e) $\frac{1}{2\sqrt{2}-1} \times \frac{2\sqrt{2}+1}{2\sqrt{2}+1} = \frac{2\sqrt{2}+1}{7}$

f) $l = r\theta = 4 \times \frac{5\pi}{18} = 3.5$

Q2a) i) $y' = e^x 2x + x^2 e^x$
 $= e^x (2x + x^2) = 2e^x (2+x)$

ii) $y = \tan x \quad y' = \sec^2 x$

b) i) $\frac{1}{3} \int \frac{3x^2+3}{x^3+3} dx = \frac{1}{3} \ln(x^3+3) + C$

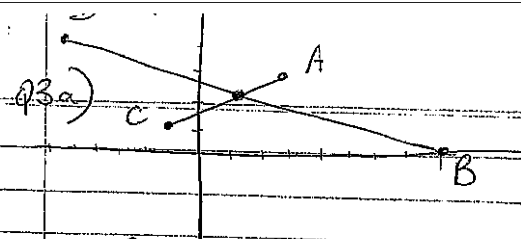
ii) $\int_1^2 \frac{2}{1-3x} dx = -\frac{2}{3} \int_{-2}^{-3} \frac{1}{1-3x} dx$
 $= -\frac{2}{3} \left[\ln(1-3x) \right]_1^2 = -\frac{2}{3} (\ln|-5| - \ln|-2|)$
 $= \frac{2}{3} \ln \frac{5}{2}$

c) $y' = -\frac{4}{x^2}$ at $x=1, y' = -4$

$y - 4 = -4(x-1)$

$y = -4x + 4 + 4$

$y = -4x + 8$



$m(AC) = \frac{1}{2}$

$m(DB) = -\frac{2}{7}$

Eqn AC

$x - 2y = -3$

Eqn DB

$y = -\frac{2x}{7} + \frac{22}{7}$

Solving Sim.

$\begin{cases} x - 2y = -3 \\ 2x + 7y = 16 \end{cases}$

$\begin{cases} 2x - 4y = -6 \\ 2x + 7y = 16 \end{cases}$

$\therefore 11y = 22 \quad y = 2, \quad x = 1 \quad (1, 2)$

b) $T_{30} = 5 + (29)6$
 $= 179$

$S_{30} = \frac{30}{2} (5 + 179)$
 $= 2760$

c) $S_{\infty} = \frac{a}{1-r} = 7 \quad \text{--- (1)}$

i) $S_2 = \frac{a(1-r^2)}{1-r} = \frac{a(1-r)(1+r)}{(1-r)} = a(1+r) = \frac{48}{7}$

$a = 7(1-r)$ sub in (2)

$7(1-r)(1+r) = \frac{48}{7}$

$49(1-r)(1+r) = 48$
 $49(1-r^2) = 48$

$49 - 49r^2 = 48$
 $1 - 49r^2 = 0$

$r^2 = \frac{1}{49} \quad r = \pm \frac{1}{7}$

ii) $a = 7(1-r)$

$a = 7\left(1 - \frac{1}{7}\right) = 7 \times \frac{6}{7}$

$a = 6$

4a) $\cos^2 \theta = \frac{1}{2}$ $\cos \theta = \pm \frac{1}{\sqrt{2}}$
 $\cos \theta = \frac{1}{\sqrt{2}}$ $\cos \theta = -\frac{1}{\sqrt{2}}$
 $\theta = 45^\circ, 315^\circ$ $\theta = 135^\circ, 225^\circ$

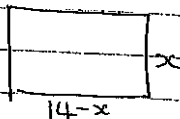
b) $m^2 + 5m - 14 = 0$
 $(m-2)(m+7) = 0$ $m = 2 \text{ or } -7$
 $x^2 = 2$ $x = \pm \sqrt{2}$
 $x^2 = -7$ no solns.

c) i) $l = r\theta$ $s = r\theta$ $\theta = \frac{s}{r}$ rads

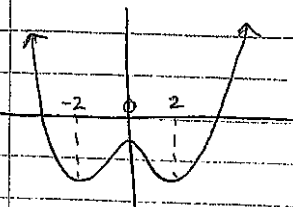
ii) $A = \frac{1}{2} r^2 (\theta - \sin \theta)$
 $= 8 \left(\frac{5}{4} - \sin \frac{5}{4} \right) = 2.408$

d) $y = \frac{1}{x^2}$ $y' = -\frac{2}{x^3}$ at $x = -1$, $y' = 2$
 Equ $y - 1 = 2(x + 1)$ $y = 2x + 3$

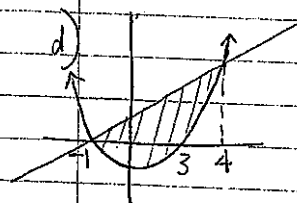
5a) $\Delta = 0$ $(-8h)^2 - 4(5k+1)(3h) = 0$
 $64h^2 - (20k+4)(3h) = 0$
 $64h^2 - 60h^2 - 12h = 0$
 $4h^2 - 12h = 0$
 $4h(h-3) = 0$
 $h = 0 \text{ or } 3$

b)  $A = x(14-x) = 14x - x^2$
 $A' = 14 - 2x$
 for sp's $A' = 0$, $14 - 2x = 0$, $x = 7$
 Max Area = $7(14-7) = 49 \text{ u}^2$

5c) $f(x) = x^4 - 8x^2 - 3$
 find sp's.



Use either to find where function is inc or dec
 $f'(x) > 0$ $x > 2$, $-2 < x < 0$
 $f'(x) < 0$ $x < -2$, $0 < x < 2$



$A = \int_{-1}^4 [(x+1) - (x^2 - 2x - 3)] dx$
 $= \int_{-1}^4 (-x^2 + 3x + 4) dx = \left[-\frac{x^3}{3} + \frac{3x^2}{2} + 4x \right]_{-1}^4$
 $= \frac{125}{6} \text{ u}^2$

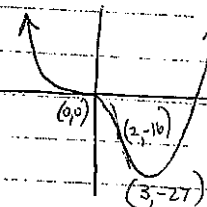
6a) $3^x = 10$ take logs $x \log 3 = \log 10$
 $x = \frac{\log 10}{\log 3} = 2.10$

b) i) $f(x) = x^4 - 4x^3$
 $f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$
 for sp's $f'(x) = 0$ $x = 0, 3$ $(0, 0)$ $(3, -27)$

ii) $f''(x) = 12x^2 - 24x = 12x(x-2)$
 for pts of inf $f''(x) = 0$, $x = 0, 2$

$\therefore (0, 0)$ horiz pt of inf.
 $(2, -16)$ non-horiz pt of inf.

At $x = 3$, $y''(x) > 0 \therefore$ Min TP at $(3, -27)$



$$6c) x^2 + 2x + (1)^2 + y^2 - 4y + (-2)^2 = 4 + 1 + 4$$

$$(x+1)^2 + (y-2)^2 = 9$$

$\therefore$ Radius 3, $(-1, 2)$ centre

$$7) y = x^2 + 6x + (3)^2 - 9 + 6$$

$$x^2 + 6x + (3)^2 = y + 9 - 6$$

$$(x+3)^2 = 1(y+3)$$

i) $\therefore$ Vertex $(-3, -3)$

$$a = \frac{1}{4}$$

ii) Focus $(-3, -2\frac{3}{4})$

iii) Dir $y = -3\frac{1}{4}$

$$b) V = \pi \int_0^2 [(4-x^2)^2 - (4-2x)^2] dx$$

$$= \pi \int_0^2 (x^4 - 12x^2 + 16x) dx$$

$$= \pi \left[\frac{x^5}{5} - 4x^3 + 8x^2 \right]_0^2$$

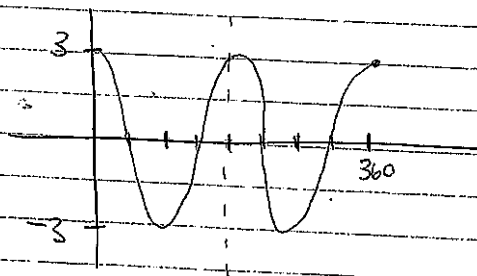
$$= \frac{32\pi}{5}$$

$$c) y = 3 \cos 2\theta$$

i) Amplitude = 3

ii) period = $\frac{2\pi}{2} = \pi$

iii)



$$8a) y = \ln\left(\frac{x-1}{x+1}\right) \quad y = \ln(x-1) - \ln(x+1)$$

$$y' = \frac{1}{x-1} - \frac{1}{x+1} \quad \text{at } x=3, y = \ln \frac{1}{2}$$

$$y' = \frac{1}{2} - \frac{1}{4} = \frac{1}{4} \quad \therefore \text{grad normal} = -4$$

$$\therefore \text{Equ of normal: } y - \ln \frac{1}{2} = -4(x-3)$$

$$y = -4x + 12 + \ln \frac{1}{2}$$

$$b) \begin{array}{|c|c|c|c|c|c|} \hline x & 1 & \frac{1}{2} & 2 & 2\frac{1}{2} & 3 \\ \hline y & e^{-2} & e^{-3} & e^{-4} & e^{-5} & e^{-6} \\ \hline \end{array}$$

$$h = \frac{1}{2}$$

$$\approx 0.067$$

$$c) 4\cos^2\theta + 3\sin\theta - 4 = 0$$

$$\text{but } \cos^2\theta = 1 - \sin^2\theta$$

$$4 - 4\sin^2\theta + 3\sin\theta - 4 = 0$$

$$4\sin^2\theta - 3\sin\theta = 0$$

$$\sin\theta(4\sin\theta - 3) = 0 \quad \therefore \sin\theta = 0 \text{ or } \sin\theta = \frac{3}{4}$$

$$\therefore \theta = 0^\circ, 180^\circ, 360^\circ$$

$$\theta = 48.6^\circ, 131.4^\circ$$

$$d) y = \sqrt{6+x} \sqrt{3-2x}$$

$$y' = v u' + u v'$$

$$= \frac{\sqrt{3-2x}}{2\sqrt{6+x}} + \frac{\sqrt{6+x}}{\sqrt{3-2x}} \cdot -1$$

$$= \frac{\sqrt{3-2x}}{2\sqrt{6+x}} - \frac{\sqrt{6+x}}{\sqrt{3-2x}}$$

$$= \frac{-12 - 2x + 3 - 2x}{2\sqrt{3-2x}\sqrt{6+x}}$$

$$= \frac{-4x-9}{2(3-2x)^{1/2}(6+x)^{1/2}}$$

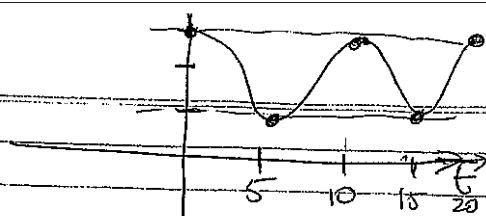
$$\left. \begin{array}{l} u = (6+x)^{\frac{1}{2}} \\ u' = \frac{1}{2}(6+x)^{-\frac{1}{2}} = \frac{1}{2\sqrt{6+x}} \end{array} \right\}$$

$$v = \sqrt{3-2x}$$

$$v' = \frac{1}{2}(3-2x)^{-\frac{1}{2}} = -\frac{1}{\sqrt{3-2x}}$$

$$= -\frac{1}{\sqrt{3-2x}}$$

Q1a) $x = 4 + 2 \cos \frac{\pi}{5} t$



i/

- Ampl = 2
- graph moves up 4. (Range $2 \leq y \leq 6$)
- period = $\frac{2\pi}{\pi/5} = 10$

ii/

$$v = \dot{x} = -\frac{2\pi}{5} \sin \frac{\pi}{5} t$$

When $v=0$, $\sin \frac{\pi}{5} t = 0$

(SPk)

$$\therefore \frac{\pi}{5} t = 0, \pi, 2\pi, 3\pi, 4\pi,$$

$$t = 0, 5, 10, 15, 20$$

b) $P = P_0 e^{kt}$

i/ When $t=15$, $P=2P_0$ $\frac{2P_0}{P_0} = e^{k \times 15}$

$$2 = e^{15k}$$

Take logs.

$$\log 2 = 15k \log e$$

$$\therefore k = \frac{\log 2}{15}$$

ii/ Find when $P = 4P_0$

$$4P_0 = P_0 e^{\frac{\log 2}{15} t}$$

Take logs

$$4 = e^{\frac{\log 2}{15} t}$$

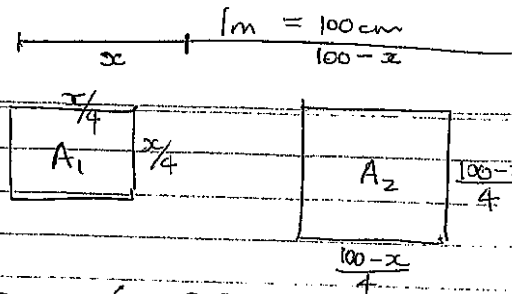
$$\log 4 = \frac{\log 2}{15} t \log e$$

$$t = \frac{15 \log 4}{\log 2} = 30 \text{ yrs}$$

iii/ $P = 30000 e^{35 \left(\frac{\log 2}{15} \right)}$

$$= 151190$$

Q1b



$$A_1 + A_2 = 325$$

$$\frac{x^2}{16} + \frac{(100-x)^2}{16} = 325$$

$$x^2 + 10000 - 200x + x^2 = 5200$$

$$2x^2 - 200x + 4800 = 0$$

$$(\pm 2) \quad x^2 - 100x + 2400 = 0$$

$$(x-40)(x-60) = 0$$

$$\therefore x = 40 \text{ or } 60$$

$$\therefore P_1 = 40 \text{ cm} \quad P_2 = 60 \text{ cm}$$

b)



$$V = 54\pi \text{ cm}^3$$

$$\text{To find } h \quad \pi r^2 h = 54\pi$$

$$h = \frac{54}{r^2}$$

i/

$$\therefore \text{Surface Area} = 2\pi r^2 + 2\pi r h$$

$$A = 2\pi r^2 + 2\pi r \times \frac{54}{r^2}$$

$$A = 2\pi r^2 + \frac{2\pi \times 54}{r} = 2\pi r^2 + \frac{108\pi}{r}$$

ii/

$$A' = 4\pi r - \frac{108\pi}{r^2}$$

For SPk $A' = 0$

$$\frac{108\pi}{r^2} = 4\pi r$$

$$r^3 = 27 \quad \therefore \underline{\underline{r=3}}$$

$$h = \frac{54}{9} = \underline{\underline{6}}$$